Supplementary for ch14

Abstract

This lecture note covers the fundamental theories of bond and stock valuation. It also introduces the basic growth model known as the Gordon Growth Model for stock valuation.

Dividends Discount Model

Let P_0 is the current market price for a stock, D_i be the total dividends paid per share of the stock during the year i, r be the real interest rate.(Note that we shouldn't use real interest rate to discount stock's cash flow, because stock have the default risk, price risk it is not a risk-free investment. However, for simplicity, we omit the risk here.)

Suppose we planned to hold the stock for 2 years. Then we would receive dividends in both year 1 and year 2 before selling the stock, as shown in the following timeline:

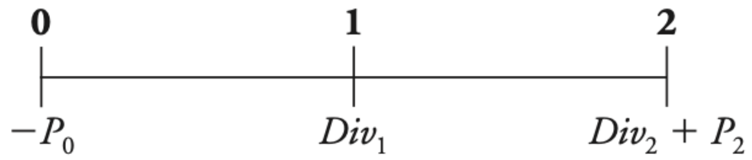

Figure 1: timeline of stock

Setting the stock price equal to the present value of the future cash flows in this case implies

$$P_0 = \frac{D_1}{(1+r)} + \frac{D_2 + P_2}{(1+r)^2}$$

We can continue this process for any number of years by replacing the final stock price with the value that the next holder of the stock would be willing to pay. Doing so leads to the general dividend-discount model for the stock price, where the horizon n is arbitrary:

$$P_0 = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \cdots + \frac{D_n}{(1+r)^n} + \frac{P_n}{(1+r)^n}$$

For the special case in which the firm eventually pays dividends and is never acquired, it is possible to hold the shares forever. Consequently, we can let n go to infinity and write the equation above as follows:

$$P_0 = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \cdots = \sum_{i=1}^{\infty} \frac{D_i}{(1+r)^i}$$

That is, the price of the stock is equal to the present value of the expected future dividends it will pay.

Example 1. Suppose Acap Corporation will pay a dividend of \$2.72 per share at the end of this year and \$2.99 per share next year. You expect Acap's stock price to be \$53.72 in two years. If Acap's equity cost of capital is 11.1%, then the price you would be willing to pay for a share of Acap stock today is

$$P_0 = \frac{2.72}{(1+0.111)} + \frac{2.99}{(1+0.111)^2} + \frac{53.72}{(1+0.111)^2} = 48.39253$$

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Gordon Growth Model

Definition 0.1. *Let the geometric series is*

$$a + ar + ar^2 + ar^3 + \cdots = \sum_{n=1}^{\infty} ar^{n-1}$$

then, the sum of geometric series is

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \qquad |r| < 1$$

The simplest forecast for the firm's future dividends states that they will grow at a constant rate, g, forever. That case yields the following timeline for the cash flows for an investor who buys the stock today and holds it:

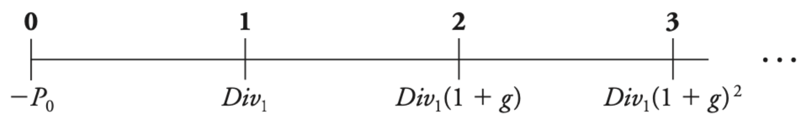

Figure 2: Constant dividend growth

Because the expected dividends are a constant growth perpetuity, we can calculate their present value as follows:

$$P_0 = \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \frac{D_0(1+g)^3}{(1+r)^3} + \cdots + \frac{D_1(1+r)^n}{(1+r)^n} + \cdots$$

By the definition 1, we can rewrite the formula

$$\begin{aligned} P_0 &= \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \frac{D_0(1+g)^3}{(1+r)^3} + \cdots \\ &= \frac{D_1}{(1+r)} + \frac{D_1(1+g)}{(1+r)^2} + \frac{D_1(1+g)^2}{(1+r)^3} + \cdots \\ &= \frac{D_1}{(1+r)} \times \frac{1}{1-\frac{1+g}{1+r}} \\ &= \frac{D_1}{(1+r)-(1+g)} \\ &= \frac{D_1}{r-g} \end{aligned}$$

Definition 0.2. *Suppose that dividends are growth at a fixed rate, g. Let P_0 is the current market price for a stock, D_1 be the total dividends paid per share of the stock during the year 1, r be the real interest rate. Then the Gordon growth model(or Constant Dividend Growth) says that*

$$P_0 = \frac{D_1}{r-g}$$

Example 2. Summit Systems will pay a dividend of \$1.46 this year. If you expect Summit's dividend to grow at a fixed rate $g = 5.1\%$ per year, what is its price per share if its equity cost of capital is 11.2%?

$D_1 = D_0(1+g) = 1.46(1+0.051) = 1.53446$

$$P_0 = \frac{D_1}{r-g} = \frac{1.53446}{0.112-0.051} = 25.15508$$

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Perpetuities

Definition 0.3. *A perpetuity is a stream of equal cash flows that occur at regular intervals and last forever.*

One example is the British government bond called a **consol** (or perpetual bond). Consol bonds promise the owner a fixed cash flow every year, forever. Using the formula for the present value, the present value of a perpetuity with payment C and interest rate r is given by

$$
\begin{aligned}
PV &= \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \cdots = \sum_{n=1}^{\infty} \frac{C}{(1+r)^n} \\
&= \frac{C}{1 - \frac{1}{1+r}} = \frac{C}{\frac{1+r-1}{1+r}} \\
&= \frac{C}{r}
\end{aligned}
$$

Example 3. NoGrowth Corporation currently pays a dividend of \$2.36 per year, and it will continue to pay this dividend forever. What is the price per share if its equity cost of capital is 13% per year?

The equity cost of capital in this case can be considered as the real interest rate. Therefore, we can derive the price is

$$P_0 = \frac{2.36}{0.13} = 18.1538$$

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Yield To Maturity

Definition 0.4. *The yield to maturity of a bond is the discount rate that sets the present value of the promised bond payments equal to the current market price of the bond.*

Example 4. Suppose that a one-year, risk-free, zero-coupon bond with a \$100,000 face value has an initial price of \$96,618.36. If you purchased this bond and held it to maturity, you would have the following cash flows:

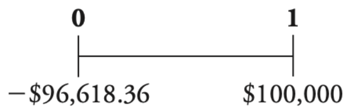

Figure 3: cash flows

According to the definition, the yield to maturity of the one-year bond solves the following equation:

$$96,618.36 = \frac{100,000}{1 + YTM}$$

In this case,

$$1 + YTM = \frac{100,000}{96,618.36} = 1.035$$

That is, the yield to maturity for this bond is 3.5%.

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Example 5. The U.S. Treasury has just issued a five-year, face value is \$100 bond with a 3% coupon rate and annual payments. The market price of this bond is 90.2853. We can calculate the YTM of this bond is

$$90.2853 = \frac{3}{(1+YTM)}+\frac{3}{(1+YTM)^2}+\frac{3}{(1+YTM)^3}+\frac{3}{(1+YTM)^4}+\frac{3}{(1+YTM)^5}+\frac{100}{(1+YTM)^5}$$

$$\Longrightarrow YTM = 5.26\%$$

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Unfortunately, there is no simple formula to solve for the yield to maturity directly. Instead, we need to use either trial-and-error or the annuity spreadsheet or the financial calculator.(I typically use the BA II Plus Professional calculator, and if you're planning to study financial management or investment, I highly recommend having one.)